Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25 September 2026
Team ID	SWTID-2026-3161
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	ACL rule configuration	Create a rule that restricts access based on a field value Edit or disable an existing rule Assign a rule to a specific table and operation (read / write / delete)
FR-2	Access request evaluation	Intercept every record access request before it reaches the record Read the current value of the governing field Compare the field value against the configured condition
FR-3	Access decision & response	Grant access when the condition is satisfied Deny access with a clear, safe reason when the condition fails
FR-4	Audit logging	Log every access attempt (granted and denied) with user, record, and timestamp Provide a searchable audit log for administrators

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Denial messages are clear and specific; rule configuration is simple enough for an administrator without scripting help.
NFR-2	Security	Only authorized administrators can create or modify ACL rules; field values are never exposed to users who are denied access.
NFR-3	Reliability	The ACL evaluates consistently on every access path; no request can bypass the rule.
NFR-4	Performance	The access decision adds no more than a few milliseconds of latency to a record request.
NFR-5	Availability	The ACL evaluation service is available whenever the platform itself is available, with no separate downtime window.
NFR-6	Scalability	Evaluation time does not noticeably degrade as the number of rules or records grows.